

PROBLEMAS 8

$$f = 2\text{GHz} = 0.5\text{ns}$$

$$R (\text{access mem}) = I + R_{\text{read}} + W_{\text{write}} = 1000000$$

$$C_{p.f_i} = \frac{21.6 + (8-1) \cdot 6.6 \text{ ns}}{0.5 \text{ ns/clock}} = 135.6$$

$$C_{p.f_{ld}} = \frac{41.2 + (8-1) \cdot 9.7}{0.5} = 218.2$$

$$C_{p.f_{sw}} = \frac{41.2 + 33.3 + (8-1) \cdot 9.7 + 33.3}{0.5} = 284.8$$

$$\begin{aligned} \textcircled{1} \quad C_{ef} &= \frac{(\#f_i \cdot C_{p.f_i}) + (\#f_{ld} \cdot C_{p.f_{ld}}) + (\#f_{sw} \cdot C_{p.f_{sw}})}{R} \\ &= \frac{7000 \cdot 135.6 + 2000 \cdot 218.2 + 6000 \cdot 284.8}{1000000} = 3.094 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad C_{P_{\text{ired}}} &= C_{P_{\text{Iuse}}} + \left(\frac{R}{I} C_{ef} \right) \\ &= 3.5 + \left(\frac{1000000}{700000} \cdot 3.094 \right) = 7.92 \end{aligned}$$

$\textcircled{3}$

$$T_{ex} = I \times CPI \times T_{ci}$$

$$T_{ex} = 700000 \cdot 7.92 \cdot 0.5 = 2772000 \text{ ns}$$